Layer 14: Electronics and Potentiostat Noise Budget

From Electrode Current to Per-Point Noise, Peak-Area Noise, and the Electronic Detection Floor

Eton iGEM 2026, Dry Lab — Layer 14 of 17-Layer Biosensor Pipeline — `noise_budget.py` · `sigma_area.py` · `snr_vs_gamma.py` · `mains_spectrum.py`

Layer 14 sits between the electrode and the peak-extraction code. It takes the electrode capacitance and background current from Layers 4–6 and the SWV waveform from Layers 11–12, and returns three things: (i) the RMS current noise $\sigma_{\Delta I}$ on every net SWV point, which Layer 12 consumes as "σ_{noise} from Layer 14"; (ii) the resulting noise σ_{area} on the fitted MB peak area, which feeds Layer 13; and (iii) a decision on whether the limit of detection is set by the electronics or by something else. The noise budget is also a design tool: it says which component to change if the electronic floor ever becomes the problem.

Scope and corrections. The reference architecture is the DStat potentiostat, as specified in the planning file (`contents.md`, Layer 14). The ProteaSign schematic (Stakeholder Document) instead shows an AD5940 analogue front end with a TMUX1104 multiplexer, so both are evaluated (Section I). Checking the planning file's arithmetic in code exposed several errors (ADC step sizes off by 10^3, electrode capacitance off by 10^1, an omitted noise source, and an inconsistent worked example). They are corrected here and tabulated in Section VIII; the qualitative conclusions of the planning file survive, but several numbers change by factors of 2–10.

I. Signal Chain, Conventions and Inputs

Hardware reference

DStat (Dryden and Wheeler, 2015 [1]) uses an ATxmega256A3U microcontroller (not a PIC, as the planning file states), an external 16-bit DAC (46 μV step on a 3 V range; 76 μV on 5 V), a 24-bit sigma-delta ADC whose digital filter has zeros at multiples of its output rate, and a transimpedance amplifier (TIA) with seven selectable feedback resistors from 100 Ω to 100 MΩ. Its measured open-circuit noise levels off near 200 fA (60 Hz sample rate, no cell attached), giving an instrument detection limit near 600 fA [1]. The ProteaSign schematic uses an AD5940 [4] (16-bit SAR ADC at 800 kSPS, 1.82 V internal reference, TIA resistor 0.2–512 kΩ per the schematic) behind a TMUX1104 4:1 multiplexer (2.5 Ω on-resistance, 1.5 pC charge injection, 3 pA leakage [3]). The DAC contributes negligibly: a 76 μV step is 3.8% of the 2 mV staircase step, and its maximum potential error (±38 μV) is 0.2% of the 20 mV peak σ_E.

Conventions

All noise values are RMS currents referred to the working-electrode input for *one* SWV current sample, taken at the end of a half-cycle of duration $\tau = 1/(2f_{\text{SWV}})$. A boxcar average over the half-cycle has noise bandwidth $B = 1/(2\tau) = f_{\text{SWV}} = 60$ Hz. (The planning file uses $B = f_{\text{SWV}}/2 = 30$ Hz, which would need an integration window longer than the half-cycle.) The net SWV current is $\Delta I_n = I_{\text{fwd},n} - I_{\text{rev},n}$; the two samples are independent, so

$$\sigma_{\Delta I} = \sqrt{2}\,\sigma_{\text{sample}} = \sqrt{2}\sqrt{\textstyle\sum_k \sigma_k^2}. \tag{I.1}$$

Inputs, all taken from upstream layers except where marked, are in Table 1.

II. Noise Sources

Johnson noise in R_f

$$i_J = \sqrt{4k_B T / R_f} \tag{II.1}$$

gives 0.406, 0.128 and 0.041 pA/$\sqrt{\text{Hz}}$ for $R_f = 100$ kΩ, 1 MΩ and 10 MΩ at 298 K (the planning file's values are confirmed). Over $B = 60$ Hz at 1 MΩ this is 0.99 pA RMS.

Op-amp noise

Two terms matter. The first is the input current noise: the OPA2333 datasheet gives $i_n = 100$ fA/$\sqrt{\text{Hz}}$ at 10 Hz [2], 5–20× larger than the 5–20 fA/$\sqrt{\text{Hz}}$ assumed in the planning file (zero-drift amplifiers trade offset

Parameter	Value	Source
True electrode area A_{true}	0.013 cm^2	Layers 4, 6
SAM double-layer capacitance C_{dl}	2 $\mu\text{F}\,\text{cm}^{-2} \Rightarrow C_{\text{cell}} = 26$ nF	Layer 6
Solution resistance R_s	7.1 kΩ	Layer 5
E_{sw}, f_{SWV}, ΔE_{step}	25 mV, 60 Hz, 2 mV	Layers 5, 11, 12
Capacitive background I_{cap}	70 nA (60–78 nA)	Layers 5, 6
Peak σ_E, window	20 mV; $\pm 3\sigma_E = 61$ points	Layer 12
MB peak area A_{MB}	0.3 nA·V (peak 5.98 nA)	Layer 13
Fc peak area A_{Fc}	1 nA·V	Layer 12 range 0.1–10; **FLAG:** mid-range assumed
Initial coverage Γ_0	2.0 $\text{pmol}\,\text{cm}^{-2}$	Layer 15
Op-amp (OPA2333)	$e_n = 55$ nV/$\sqrt{\text{Hz}}$, $i_n = 100$ fA/$\sqrt{\text{Hz}}$, GBW 350 kHz	TI datasheet [2]
ADC	24-bit sigma-delta, 20 ENOB, ±2.5 V	planning file, DStat
Temperature (instrument)	298.15 K	—

Table 1: Inputs to the noise budget.

and $1/f$ performance for higher current noise). Over 60 Hz this is 0.77 pA, comparable to Johnson noise at 1 MΩ, not negligible.

The second term is absent from the planning file. The WE is held at virtual ground, so the op-amp's input voltage noise e_n appears across the electrode impedance $\approx 1/(2\pi f C_{\text{cell}})$ (valid for $f \ll 1/(2\pi R_s C_{\text{cell}}) \approx 860$ Hz) and drives a current $i(f) = 2\pi f\, C_{\text{cell}}\, e_n$ in the measurement. Integrating over the noise bandwidth,

$$i_{e_n C,\text{rms}} = 2\pi\, C_{\text{cell}}\, e_n \sqrt{B^3/3} = 2.41\,\text{pA}, \tag{II.2}$$

using $C_{\text{cell}} = 26$ nF and $e_n = 55$ nV/$\sqrt{\text{Hz}}$ (1 kHz datasheet value). This is the *largest* single term in the budget. It grows as $B^{3/2}$, independent of R_f, and is the reason raising R_f beyond a few MΩ buys almost nothing (Section III).

Shot noise of the DC background

$i_{sh} = \sqrt{2qI_{\text{cap}}} = 0.150$ pA/$\sqrt{\text{Hz}}$ for $I_{\text{cap}} = 70$ nA, i.e. 1.16 pA over 60 Hz (the planning file's density is confirmed; its summary table entry of "0.025 at 30 Hz" is inconsistent with it). Strictly, shot noise arises from carriers crossing a barrier; a capacitive charging current does not have it, so this term is a conservative upper bound. The true shot noise is set by the much smaller Faradaic leakage and O_2 background.

ADC quantisation

With input-referred step LSB $= 2V_{\text{ref}}/(R_f\, 2^{\text{ENOB}})$ and noise LSB/$\sqrt{12}$: at $R_f = 100$ kΩ and ±2.5 V the 20-ENOB step is 47.7 pA (48 *pA*, not the planning file's 48 nA), at 1 MΩ it is 4.8 pA, giving 1.38 pA RMS. A nominal 24-bit step at 100 kΩ is 3 pA (not 3 nA). The planning file's lesson survives with correct numbers: at 100 kΩ quantisation dominates (14 pA RMS against ≈3 pA of analogue noise), so R_f must be scaled up to use the ADC range. The DStat paper reaches the same conclusion: for any ADC up to 16 bits, the step, not the analogue noise, limits resolution at every gain [1] (Fig. 1C).

1/f noise

The OPA2333 is chopper-stabilised, so the corner is negligible. Integrating $i_n^2 f_c/f$ with $f_c = 0.1$ Hz from 0.1 to 60 Hz gives 0.08 pA. Low-frequency drift is also suppressed by the forward–reverse differencing and removed by the baseline fit (Layer 12).

TIA bandwidth and stability

The planning file's $C_f = 1$–10 pF (159 kHz at 100 kΩ) ignores the 26 nF source capacitance, which destabilises a TIA. Phase margin requires $C_f \gtrsim \sqrt{C_{\text{in}}/(2\pi R_f\,\text{GBW})} = 109$ pF at 1 MΩ, so the signal bandwidth is $1/(2\pi R_f C_f) = 1.46$ kHz, not 15.9 kHz. This is still 24× above f_{SWV}, so the conclusion is unchanged, but the component value must be corrected.

Budget

Table 2 and Fig. 1 give the base case (DStat, $R_f = 1$ MΩ, $f_{\text{SWV}} = 60$ Hz), and the same model for the AD5940 case. The AD5940 column uses $R_f = 512$ kΩ, 16 bits, a 1.82 V span and no PGA; **FLAG:** its amplifier

noise is a placeholder (OPA2333 values) and the ADC span is approximate. The chip's PGA (gain 1–9) and 800 kSPS oversampling would reduce the ADC term and are not credited.

Source	Density (pA/$\sqrt{\text{Hz}}$)	DStat RMS/sample (pA)	AD5940 case (pA)
Johnson (R_f)	0.128 (1 MΩ)	0.99	1.39
Op-amp i_n	0.100	0.77	0.77
Op-amp $e_n \times C_{\text{cell}}$	0.539 (at 60 Hz)	**2.41**	2.41
Shot (I_{cap}, upper bound)	0.150	1.16	1.16
ADC quantisation	—	1.38	**15.66**
$1/f$	—	0.08	0.08
Total per sample		3.26	15.96
Total per net SWV point ($\times\sqrt{2}$)		**4.61**	22.58

Table 2: Noise budget, $B = 60$ Hz, $C_{\text{cell}} = 26$ nF. Without the e_nC term (the planning-file model) the DStat total is 3.11 pA per net point.

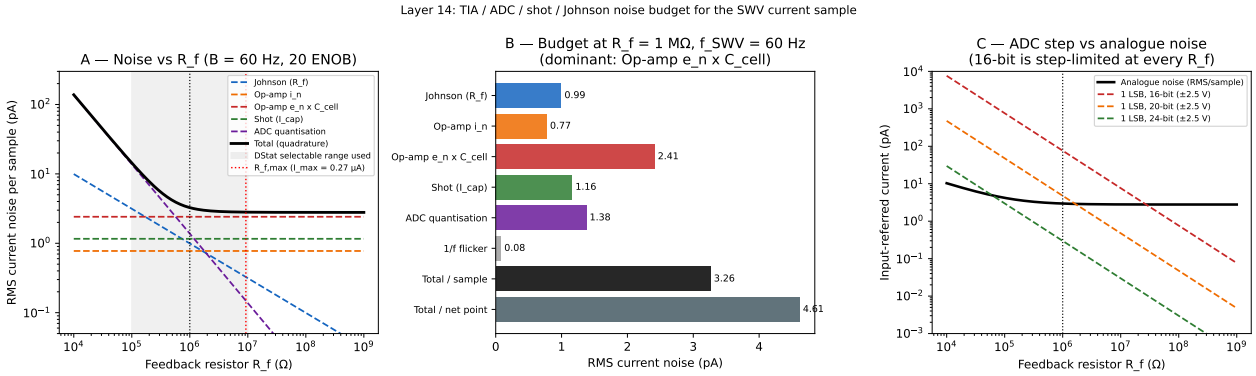


Figure 1: A: current noise per sample versus R_f for each source. The flat terms (e_nC, i_n, shot) set a floor near 2.7 pA; the ADC term dominates below ≈300 kΩ. B: budget at 1 MΩ. C: ADC step versus analogue noise; the 16-bit step exceeds analogue noise at every R_f.

III. Choice of R_f

R_f	$\sigma_{\Delta I}$ (pA)	Input range	Stable C_f	TIA BW	Dominant term
100 kΩ	20.4	±25 µA	344 pF	4.6 kHz	ADC quantisation
1 MΩ	4.61	±2.5 µA	109 pF	1.46 kHz	e_nC
10 MΩ	3.97	±0.25 µA	34 pF	463 Hz	e_nC

Table 3: R_f trade-off (DStat, 20 ENOB).

Going from 100 kΩ to 1 MΩ cuts the net noise 4.4×; going on to 10 MΩ cuts it only a further 14%, because the R_f-independent terms dominate, and it would clip if the MB peak is as large as Layer 11's 0.27 µA ($R_{f,\max} = 2.5$ V/0.27 µA = 9.3 MΩ). The planning file's recommendation of $R_f = \mathbf{1\ M\Omega}$ is therefore confirmed, now on the basis of range rather than noise: it is the largest decade that safely holds the biggest plausible peak. If Layer 13's smaller peak is right, autoranging to 10 MΩ is possible but unnecessary.

IV. Multiplexer Settling and Charge Injection

When the multiplexer connects WE_i, the electrode charges through $R_{\text{mux}} + R_s$:

$$\tau_{\text{settle}} = (R_{\text{mux}} + R_s)\, C_{\text{cell}} = (2.5 + 7100)\,\Omega \times 26\ \text{nF} = 185\ \mu\text{s}. \tag{IV.1}$$

The planning file's 14 µs used $C = 2$ nF, ten times too small; the wait rule "$10\tau \approx 140$ µs" becomes 1.85 ms. The schematic leaves idle electrodes floating, so on reconnection the electrode may sit several hundred mV

from E_{start}. For a 0.3 V step the initial current is $0.3/7102 = 42$ μA, which saturates the 2.5 μA range at 1 MΩ, and it takes $\ln(42\ \mu\text{A}/1\ \text{pA}) = 17.6\tau = 3.2$ ms to decay below 1 pA (half-cycle at 60 Hz: 8.3 ms). Charge injection is negligible for the TMUX1104: 1.5 pC on 26 nF is a 58 μV step (2.9% of one 2 mV step). The planning file's ADG1606 value (10–100 pC) would give up to 3.8 mV, larger than a staircase step. (The TMUX1104 is the part in the schematic; ADG1606 is a 16:1 device.)

Firmware requirement (replaces ">100 μs"): wait $\geq 20\tau \approx 3.7$ ms after every mux switch before the first sample, *and* hold at E_{start} for $5\tau_{\text{relax}}$ (Layer 11, Sec. IX). The second condition ($\tau_{\text{relax}} = 1/k_{\text{tot}} \sim$ 10–100 ms for $k^0 \sim$ 10–100 s^{-1}) dominates the wait. A 206-step MB-window scan at 60 Hz takes 3.4 s per electrode. **FLAG:** TIA recovery time from the initial saturation is not modelled and should be measured.

V. Mains Interference

Amplitude

Mains couples in as a displacement current through a stray capacitance C_c from the wiring to the high-impedance WE loop, reduced by a shield of effectiveness SE:

$$i_{pk} = \frac{2\pi f_m\, C_c\, \sqrt{2}\, V_{\text{rms}}}{10^{\,\text{SE}/20}}. \tag{V.1}$$

For a UK grid (230 V, 50 Hz), $C_c = 1$ pF gives 102 nA, reproducing the planning file's "1–100 nA" unshielded range. **FLAG:** C_c and SE are scenario assumptions, not measurements; they must be measured by shorting the WE input and recording with the cell disconnected. (The planning file quotes the shielded level as "<0.1 nA" in the text but "<0.1 pA/$\sqrt{\text{Hz}}$" in its table; a tone is not a noise density, so amplitude in nA is the right description.)

Aliasing into the SWV waveform

Mains is not white noise here. The net current is the difference of two samples spaced τ apart, once per staircase step. A tone $A\cos(2\pi f_m t + \phi)$ therefore appears as $\Delta I_n = G\,A\sin(2\pi\nu n + \phi')$ with

$$G = 2\,|\sin(\pi f_m \tau)|\ \,|\text{sinc}(f_m T_{\text{int}})|\,, \qquad \nu = (f_m/f_{\text{SWV}}) \bmod 1, \tag{V.2}$$

where T_{int} is the averaging window of each sample and $\text{sinc}\,x = \sin\pi x/\pi x$. The tone becomes a ripple in E-space of period $\Delta E_{\text{step}}/\nu$, which the Savitzky–Golay filter, cubic baseline and Gaussian fit turn into an area error. For $T_{\text{int}} = \tau$, $G = 2\sin^2(\pi f_m\tau)/(\pi f_m \tau)$. Table 4 lists the gain and beat for a 50 Hz grid.

f_{SWV} (Hz)	25	30	45	60	80	100
Gain G	0.000	0.19	1.11	1.43	1.41	1.27
Alias (Hz)	0	10	5	10	30	50
Beat period in E (mV)	DC	6.0	18	12	5.3	4.0

Table 4: Transfer of a 50 Hz tone into the net SWV current ($T_{\text{int}} = \tau$). On a 60 Hz grid the null is at $f_{\text{SWV}} = 30$ Hz.

Effect on the peak area

Monte Carlo through the Layer 12 chain (random phase, ±0.2 Hz grid detuning, tone amplitude in the linear regime) gives an area error of **0.98% per nA of pickup** at $f_{\text{SWV}} = 60$ Hz on a 50 Hz grid (linear to within 3% from 10 to 300 pA). Table 5 converts the scenarios.

Scenario	Pickup	% of MB peak	σ_A/A at 60 Hz	σ_A/A at 45 Hz
Unshielded, 1 pF	102 nA	1710	>100% (unrecoverable)	>100%
Unshielded, 0.1 pF	10.2 nA	171	>100%	>100%
Grounded foil, 30 dB	0.323 nA	5.4	0.30%	2.4%
Closed cage, 60 dB	10.2 pA	0.17	0.010%	0.085%
Electronic floor (Section VI): $\sigma_A/A = 0.037\%$; Layer 13 budget: 3%				

Table 5: Mains-induced area error, 50 Hz grid.

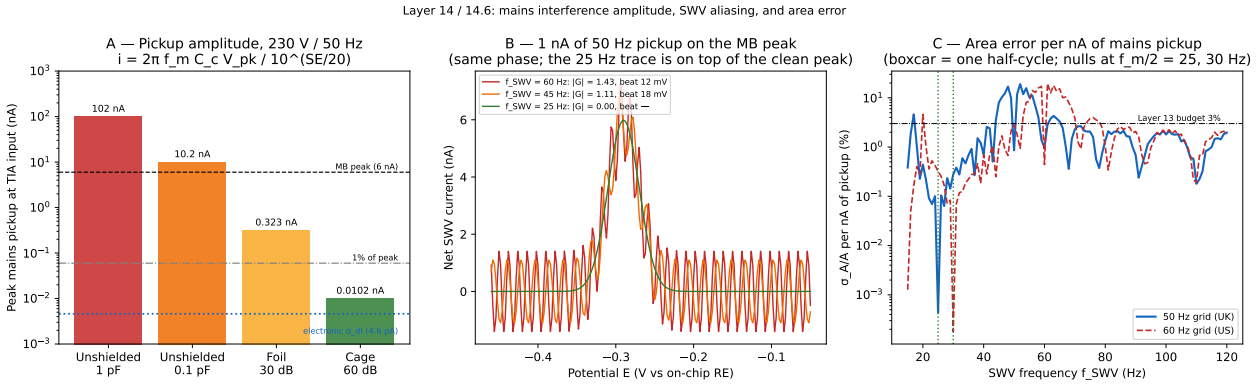


Figure 2: A: pickup amplitude by scenario. B: 1 nA of 50 Hz pickup on the MB peak at three SWV frequencies (25 Hz is synchronous and leaves the peak clean). C: area error per nA versus f_{SWV} for 50 and 60 Hz grids; the response is erratic, with deep nulls only at $f_m/2$.

Design consequences

1. **Shielding is non-negotiable.** Keeping the area error below 3% at 60 Hz needs ≈31 dB, below 1% needs ≈40 dB, and pushing it below the electronic floor needs ≈69 dB from the 1 pF open-bench case. A closed grounded cage (60 dB assumed) is sufficient.
2. **Merely detuning f_{SWV} from the mains frequency does not help** (the planning file's option 2). It moves the ripple in E-space; on a 50 Hz grid, 45 Hz is about 8× *worse* than 60 Hz because the 18 mV beat is close to the peak width.
3. **Synchronous integration works.** If each half-cycle spans exactly one mains period ($\tau = 1/f_m$, i.e. $f_{\mathrm{SWV}} = f_m/2 = 25$ Hz UK, 30 Hz US), both the differencing and boxcar factors vanish and the tone and its harmonics are nulled. This needs the mux/ADC timing to be locked to $\tau = 20$ ms and is available only if k^0_{ET} (Layers 10–11) puts f_{opt} at or below this range. Layer 11 quotes $f_{\mathrm{opt}} \approx k^0/2 \approx 15$ Hz for $k^0 = 30\ \mathrm{s}^{-1}$, which is consistent.
4. The DStat sigma-delta filter offers the same effect through its zeros at multiples of the output rate [1]; the AD5940's sinc^3/sinc^2 filter should be configured equivalently (**FLAG:** to be checked against the datasheet).

VI. Propagation to the Peak Area

Layer 12 fits $\Delta I^{\mathrm{corr}}(E) = A_{\mathrm{peak}} \exp[-(E - E_0)^2/2\sigma_E^2]$ over $\pm 3\sigma_E$ (61 points at 2 mV), with area $A_{\mathrm{area}} = A_{\mathrm{peak}}\sigma_E\sqrt{2\pi}$ and variance from its Eq. (IV.3). For white noise $\sigma_{\Delta I}$ on each point every term of that variance scales as $\sigma^2_{\Delta I}$ and is independent of A_{peak}, so σ_{area} does not depend on Γ. Five estimates for $A_{\mathrm{MB}} = 0.3$ nA·V (peak 5.98 nA) and $\sigma_{\Delta I} = 4.61$ pA are in Table 6.

Estimator	σ_{area} (fA·V)	SNR	σ_A/A (%)
Direct integration $\sigma_{\Delta I}\sqrt{N}\,\Delta E$ (planning file)	72.1	4160	0.0240
Free-shape Gaussian fit, Cramér–Rao (Layer 12 Eq. IV.3)	67.3	4450	0.0224
Fixed-shape fit (E_0, σ_E locked), Cramér–Rao	55.0	5460	0.0183
Full Layer 12 pipeline, Monte Carlo (SG → baseline → fit)	**112.5**	**2670**	**0.0375**
Windowed sum after baseline subtraction, Monte Carlo	118.7	2530	0.0396

Table 6: Peak-area noise from electronics alone (3000 Monte Carlo trials). The full-pipeline value is adopted as the Layer 14 output.

The full pipeline is 1.7× worse than the Cramér–Rao bound, which counts neither the variance of the fitted cubic baseline nor the correlation introduced by smoothing; this is why the Monte Carlo number is adopted. The amplitude–width correlation is −0.58, confirming Layer 12's remark that the covariance term must not be dropped. Three further points follow. (i) The Gaussian fit is only 5% quieter than a plain windowed sum, so the planning file's explanation that the fit "extracts more information than a peak-height measurement" is not a noise advantage of the fit. The larger gain is from locking the peak shape (2× against the pipeline, though this gap mixes baseline and smoothing effects). (ii) $\sigma_{\mathrm{area}} \propto \sqrt{\Delta E_{\mathrm{step}}}$ (Fig. 3C): a 1 mV step lowers it by 30% at the price of doubling the scan time. (iii) The pipeline has a −0.27% smoothing bias, common to $R(T)$ and

$R(0)$ and therefore cancelled in ρ. The Fc peak (1 nA·V, 121 points) has a relative electronic noise of only 0.016%. Layer 12's "signal below LOD" flag corresponds to $A_{\text{peak}} < 3\sigma_{\Delta I}/\sqrt{N_{\text{pts}}} = 1.8$ pA here, three orders of magnitude below the peak. Layer 12's quoted Savitzky–Golay [5] sum $\sum c_k^2 = 0.233$ evaluates to 0.2075 (noise factor 0.455, not 0.48).

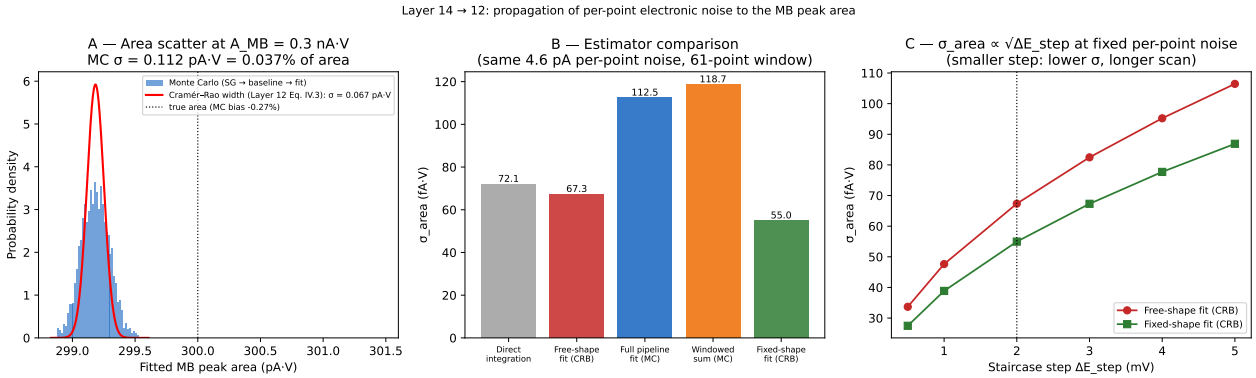


Figure 3: A: Monte Carlo scatter of the fitted area against the Cramér–Rao width. B: estimator comparison. C: dependence on the staircase step.

VII. SNR versus Probe Coverage and the Electronic Floor

In the reversible surface-confined limit, integrating Layer 11 Eq. (V.5) over the sech^2 peak gives

$$A_{\text{MB}}(\Gamma) = 2\, n_e F\, A_{\text{true}}\, \Gamma\, E_{sw} f_{\text{SWV}}\, f_{\text{kin}}, \tag{VII.1}$$

so SNR $= A_{\text{MB}}/\sigma_{\text{area}}$ is linear in Γ. The kinetic factor f_{kin} depends on k^0_{ET}, which is unmeasured, so it is inferred by requiring $A_{\text{MB}}(\Gamma_0 = 2\ \text{pmol cm}^{-2}) = 0.3$ nA·V, giving $f_{\text{kin}} = 0.020$ (the reversible limit, 15 nA·V, would give a 50× higher SNR).

Propagating electronic area noise through Layer 13 (Eqs. I.4, II.4, III.5, with $n_0 = 3$ baseline scans and an intact control electrode) gives an electronic $\sigma_{F_k}(F_k = 0)$ of **0.065%** for one electrode pair and 0.033% for four. The corresponding electronic $F_{\text{LOD}} = 3\sigma_{F_k}$ is 0.20%, roughly 100× below Layer 13's $F_{\text{min}} = 20\%$ (Fig. 4B). Unlike Layer 13's $(1 - F_k) \times 6.8\%$, the electronic σ_{F_k} stays flat near 0.04% out to $F_k = 0.99$, because the shrinking peak is measured against constant absolute noise. Layer 13's $F_{\text{max}} = 95\%$ is therefore not electronic either: the remaining probe would still have SNR ≈ 3 at $F_k = 0.9989$.

Case	SNR at Γ_0	Γ_{det} (SNR=3)	Γ (σ_A/A=3%)	$\Gamma_{0,\text{min}}$ ($\sigma_F < 1.7\%$)	Margin
DStat, full pipeline	2670	0.0022	0.025	**0.072**	28×
DStat, fixed shape	5460	0.0011	0.012	0.035	57×
AD5940 case, full pipeline	545	0.011	0.122	0.352	5.7×

Table 7: Coverage thresholds in pmol cm^{-2}. $\Gamma_{0,\text{min}}$ is the starting coverage at which the electronic part of $\sigma_{F_k}(0)$ reaches 1.7% (the requirement for $F_{\text{LOD}} = 5\%$).

Is the LOD electronics-limited?

No. Even the pessimistic AD5940 case (16-bit, no PGA, no oversampling) has SNR ≈ 550 at Γ_0, well above the 33 needed for 3%. The electronic floor (0.04%) is 80× below Layer 13's assumed 3% per-scan area precision. With mains shielded, the mains term adds only 0.01% (cage) to 0.3% (partial foil). The 3% therefore has to come from effects that Layer 14 does not model: baseline-model error, peak drift out of the fit window, differential fouling, and incomplete redox re-equilibration between scans are the candidates, but this layer does not establish which. The consequence for design is that improving the electronics (higher-resolution ADC, lower-noise op-amp) will not lower the LOD; the levers are upstream (Layers 1, 9, 12) or a measurement of the actual scan-to-scan reproducibility. This also agrees with the planning file's own conclusion, but its figure ($F_{\text{LOD}} \approx 0.02\%$) is an electronics-only floor and should not be quoted as the device LOD; the planning file's citation of a "0.6% LOD in Layer 13" does not match Layer 13, which gives 20%.

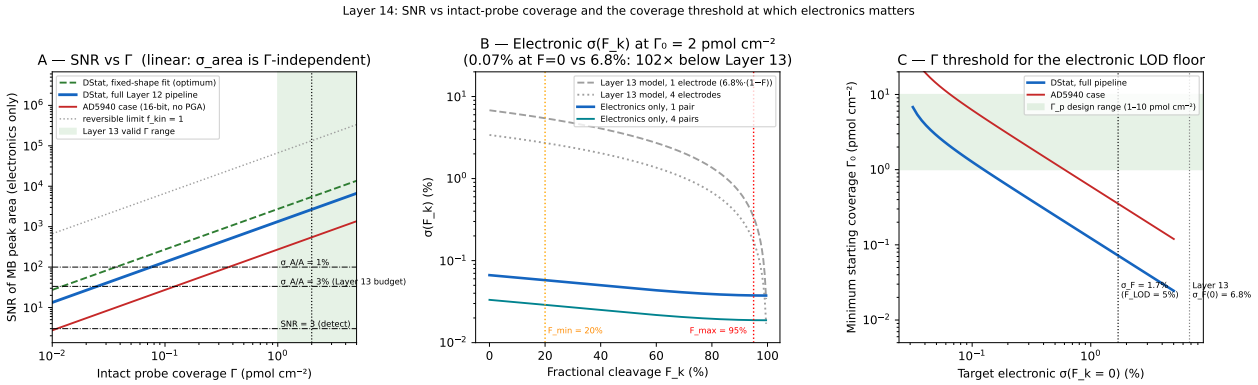


Figure 4: A: SNR versus Γ for the DStat (pipeline and fixed shape), the AD5940 case and the reversible limit. B: electronic σ_{F_k} versus F_k at $\Gamma_0 = 2$ pmol cm^{-2} against Layer 13's model. C: minimum starting coverage for a target electronic $\sigma_{F_k}(0)$.

VIII. Reconciliation with the Planning File

Quantity	`contents.md`	This report	Reason
ADC step (24 bit, ±2.5 V, 100 kΩ)	3 nA; 48 nA at 20 ENOB	3.0 pA; 47.7 pA	10^3 arithmetic slip
Op-amp current noise	5–20 fA/$\sqrt{\text{Hz}}$	100 fA/$\sqrt{\text{Hz}}$	OPA2333 datasheet
$e_n C_{\text{cell}}$ term	absent	2.41 pA (dominant)	26 nF source capacitance
TIA C_f, bandwidth	1–10 pF; 15.9 kHz	≈109 pF; 1.46 kHz	source-capacitance stability
$C_{dl}A$ for settling	2 nF	26 nF	2 μF cm^{-2} × 0.013 cm^2
τ_{settle}; wait	14 μs; >100 μs	185 μs; ≥3.7 ms + 5τ_{relax}	follows from C; floating-electrode step
Charge injection	10–100 pC (ADG1606)	1.5 pC (TMUX1104)	part in schematic
Noise per point (14.8)	2.3 pA	3.26 pA (sample); 4.61 pA (net)	2.3 pA used 100 kΩ Johnson; net point ×$\sqrt{2}$; $B = 60$ Hz
σ_{area}	26.5 fA·V (table: 2–5)	67 (CR); 112 (pipeline)	table and text disagree by 10×
SNR at A_{MB}	30,600 (0.81 nA·V)	2670 (0.3 nA·V, pipeline); 12,000 at 0.81 nA·V (CR)	noise and amplitude differences
Mains shielded level	<0.1 nA (text) vs pA/$\sqrt{\text{Hz}}$ (table)	10 pA tone	tone, not density
Detune f_{SWV} to avoid mains	suggested 45 Hz	8× worse than 60 Hz	alias beat near peak width
Microcontroller	PIC or similar	ATxmega256A3U	DStat paper

Table 8: Corrections to the planning file.

IX. Key Outputs

Output	Value	Feeds
$\sigma_{\Delta I}$ per net SWV point	4.61 pA (DStat, 1 MΩ, 60 Hz); 22.6 pA (AD5940 case)	Layer 12 (σ_{noise})
σ_{area} (MB, full pipeline)	112 fA·V = 0.037% of 0.3 nA·V	Layers 12, 13
Electronic $\sigma_{F_k}(0)$	0.065% (1 pair), 0.033% (4 pairs)	Layers 13, 15
SNR at Γ_0	2670 (DStat); 545 (AD5940 case)	LOD is not electronics-limited
Minimum Γ_0 for electronic $\sigma_F < 1.7\%$	0.072 pmol cm^{-2} (28× margin)	Layers 7, 9
R_f, C_f	1 MΩ, ≈110 pF	Hardware spec
Mux wait	≥ 3.7 ms + hold 5τ_{relax}	Firmware spec
Shielding	Closed grounded cage; ≈40 dB for <1% at 60 Hz	Hardware spec
f_{SWV}	Avoid 45 Hz; 25 Hz (UK) synchronous if k^0 allows	Layers 10, 11

X. Limitations

1. **Hardware is not fixed.** The ProteaSign schematic shows an AD5940, but the reference model is DStat. The AD5940 case uses placeholder amplifier noise, an approximate ADC span, and ignores its PGA and oversampling (pessimistic).
2. e_n **is treated as white** at its 1 kHz value; chopper ripple and the op-amp's noise gain at higher frequencies are not modelled.
3. **Shot noise is an upper bound** because the background is capacitive charging current.
4. **Signal amplitude is uncertain by up to two orders of magnitude.** Layer 13 uses $A_{\mathrm{MB}} = 0.3$ nA·V; the planning file 0.81; Layer 11's Eq. (V.5) as written evaluates to 10.8 μA at $\Gamma = 2 \times 10^{-10}$ (108 nA at 2×10^{-12} mol cm^{-2}) against its quoted 0.27 μA. The SNR spans 4×10^3 to 2×10^5 over these choices, so the conclusion is robust, but f_{kin} and A_{Fc} need measurement.
5. **Mains scenarios are assumed**, not measured: single sinusoid, no harmonics or magnetic pickup, idealised boxcar, 0.2 Hz detuning.
6. **White, independent noise** is assumed between steps. Correlated drift, baseline-model error and other systematics are excluded, so the electronic σ_F is a floor, not a prediction of the achieved σ_F.
7. The electronic σ_F assumes an intact control electrode ($\rho_c = 1$), $n_0 = 3$ baseline scans and no background cleavage.
8. Floating-electrode reconnection saturates the TIA (42 μA against a 2.5 μA range); recovery time is unmodelled.
9. Layer 11 Sec. VI's residual capacitive current (2.4 pA) appears to be a unit slip for 2.4 nA when its own inputs are used; worth checking because this feeds the baseline.

References

[1] Dryden MDM, Wheeler AR (2015). DStat: a versatile, open-source potentiostat for electroanalysis and integration. *PLoS ONE* 10(10):e0140349. DOI: 10.1371/journal.pone.0140349. – microcontroller, 16-bit DAC, 24-bit sigma-delta ADC, TIA ranges, measured open-circuit noise, ADC-step limitation, mains rejection by filter zeros.

[2] Texas Instruments. OPA333/OPA2333 1.8-V micropower CMOS operational amplifiers, zero-drift series, datasheet SBOS351. – $e_n = 55$ nV/$\sqrt{\mathrm{Hz}}$ (1 kHz), $i_n = 100$ fA/$\sqrt{\mathrm{Hz}}$ (10 Hz), GBW 350 kHz.

[3] Texas Instruments. TMUX1104 5-V, low-leakage-current, 4:1 precision multiplexer, datasheet SCDS392. – R_{on} 2–2.5 Ω, charge injection 1.5 pC, leakage 3 pA.

[4] Analog Devices. AD5940/AD5941 high precision, impedance and electrochemical front end, datasheet. – 16-bit 800 kSPS SAR ADC, 1.82 V reference, TIA current range 50 pA to 3 mA. (Only headline specifications were consulted.)

[5] Savitzky A, Golay MJE (1964). Smoothing and differentiation of data by simplified least squares procedures. *Anal Chem* 36:1627–1639. DOI: 10.1021/ac60214a047. – as used in Layer 12; $\sum c_k^2$ recomputed numerically.

[6] Horowitz P, Hill W. *The Art of Electronics*, Ch. 8; and Bard AJ, Faulkner LR. *Electrochemical Methods*, App. A. – cited by the planning file; **not consulted directly here**. The source-capacitance noise-gain term and the Schottky formula are standard results used from first principles.

Web-accessed sources (DStat, datasheets) should be saved as PDFs in `electrochemical_device/papers/layer_14/`, following the convention of earlier layers.